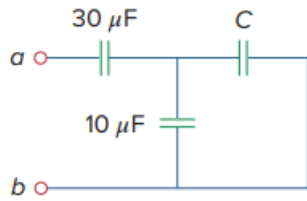


Problem

The equivalent capacitance at terminals a - b in the circuit of Fig. 6.50 is $20\ \mu\text{F}$. Calculate the value of C .

Fig. 6.50:



Step-by-step solution

Step 1 of 2

Consider the circuit in Figure 6.50 in the textbook.

In the circuit the capacitors C and $10\ \mu\text{F}$ are in parallel. Hence the equivalent capacitance is equal to $(10\ \mu\text{F} + C)$.

Redraw the circuit as shown in Figure 1.

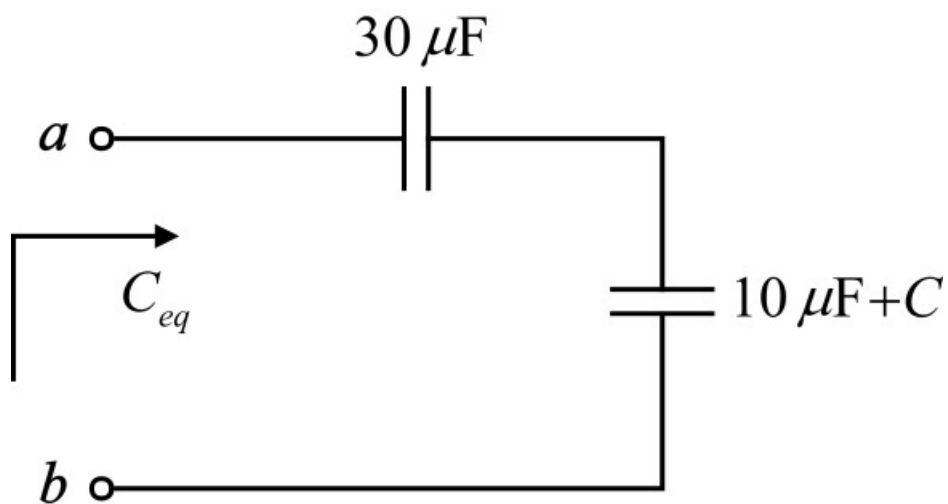


Figure 1

Step 2 of 2

The capacitors $(10\ \mu\text{F} + C)$ and $30\ \mu\text{F}$ are in series.

Hence, the total equivalent capacitance of the circuit is,

$$C_{eq} = \frac{(10\ \mu\text{F} + C)30\ \mu\text{F}}{(10\ \mu\text{F} + C) + 30\ \mu\text{F}}$$

Substitute $20\ \mu\text{F}$ for C_{eq} .

$$20\ \mu\text{F} = \frac{(10\ \mu\text{F} + C)30\ \mu\text{F}}{(10\ \mu\text{F} + C) + 30\ \mu\text{F}}$$

$$(10\ \mu\text{F} + C) + 30\ \mu\text{F} = (10\ \mu\text{F} + C)1.5$$

$$40\ \mu\text{F} + C = 15\ \mu\text{F} + 1.5C$$

$$25\ \mu\text{F} = 0.5C$$

$$C = 50\ \mu\text{F}$$

Thus, the value of unknown capacitor C is $50\ \mu\text{F}$.